

## **Phase 4: Core Development & Implementation (Weeks 13–16)**

### **Objectives**

- Implement smart contracts and integrate cross-chain functionality.
- Deploy to local testnet.
- Address any redesigns identified from earlier research.

### **Development Tasks**

- **Weeks 13–14: Contract Implementation**
  - Develop SubscriptionManager.rs (state management, payments, verification).
  - Implement PayPerView.rs (one-time access, tokens).
  - Create PurchaseVerification.rs (ownership transfers, resale).
  - Write unit tests (80% coverage) and deploy to local testnet.
- **Week 15: XCM & Integration**
  - Implement cross-chain message handlers and parachain transfers.
  - Integrate Ethereum bridge.
  - Develop chain extensions for efficiency.
  - Test initial integrations.
- **Week 16: Full System Integration**
  - Integrate all contracts.
  - Create end-to-end test scenarios.
  - Deploy the complete system to the local testnet.
  - Begin preliminary performance logging.

### **Writing Tasks (Target: 1,500 words)**

- Implementation details section draft (1,500 words).

### **Deliverables**

- Three functional smart contracts.
- Integrated local testnet deployment.
- XCM implementation.
- Unit test suite (80%+ coverage).
- Initial implementation documentation.